Econometrics: Advanced Methods

Short lecture notes for Research Metrics 2 (MQE, LMU Munich)

Derya Uysal

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Preface

These notes accompany the course *Econometrics: Advanced Methods (Research Metrics 2)* in the Munich Graduate School of Economics MQE programme. They are deliberately short. Each chapter states the key definitions and results, works through one or two examples in detail, and ends with a computer tutorial in R. Proofs are sketched where the argument is instructive and otherwise referenced.

The notes complement the lecture slides; they do not replace the textbooks listed below.

What the course covers

The objective is to introduce fundamental estimation methods in econometrics. We spend a fair amount of time on the theoretical properties of the estimators and also discuss practical aspects of implementation. The course is not a course on causality or identification, and it is not an R course, although both appear throughout.

Chapter	Topic	Computer tutorial
1	Maximum likelihood	CT 1: logit by hand, PML vs. ML
2	Method of moments and GMM, including IV	CT 2: GMM estimation of an ATE
3	Simulation-based estimation	CT 3: Monte Carlo integration, Halton draws, quadrature
4	Nonparametric density and regression estimation	CT 4: kernel density, local linear regression
5	Introduction to statistical learning	CT 5: ridge, lasso, elastic net
A	Large-sample asymptotics (assumed known; for reference)	
B	A short R primer	CT 0

Literature

- Hansen, B. E. (2022). *Econometrics*. Princeton University Press (Hansen 2022).
- Cameron, A. C. and Trivedi, P. K. (2005). *Microeconometrics: Methods and Applications*. Cambridge University Press (Cameron and Trivedi 2005).
- Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. MIT Press (Wooldridge 2010).
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2023). *An Introduction to Statistical Learning*, 2nd ed. Springer (James et al. 2023).

- Casella, G. and Berger, R. L. (2002). *Statistical Inference*, 2nd ed. Duxbury (Casella and Berger 2002).

Chapter-specific readings are given at the start of each chapter.

Software

All tutorials use R. Install a recent version of R and an editor (RStudio or Positron), then install the packages used in the notes:

```
install.packages(c("sandwich", "lmtest", "MASS", "nleqslv",
                   "randtoolbox", "statmod", "glmnet", "haven"))
```

Every tutorial can be run from top to bottom. Data sets are in the `data/` folder of the course material on Moodle.

How to use the tutorials

Each tutorial first states the problems. Try them yourself before opening the solution boxes. Solutions are released after the tutorial session.

Conventions

Vectors and matrices are in bold, $\boldsymbol{\theta}$ denotes a generic parameter vector and $\boldsymbol{\theta}_0$ its true value. ln is the natural logarithm. $\xrightarrow{p}$ and $\xrightarrow{d}$ denote convergence in probability and in distribution.

Part I

Estimation principles

1 Maximum Likelihood

> **Readings**
>
> Wooldridge (2010), Ch. 13.1–13.7 and 13.11; Cameron and Trivedi (2005), Ch. 5.6–5.7; Hansen (2022), Ch. 10. Applications: Smith (2009) and Winkelmann (2004).

Maximum likelihood (ML) is the workhorse estimator for fully specified parametric models. This chapter introduces the likelihood principle, derives the large-sample properties of the MLE, discusses what happens when the assumed density is wrong, and closes with the classical trinity of tests.

1.1 Parametric models

We first consider unconditional distributions. A **parametric model** for a random variable Y is a complete probability function $f(y \mid \boldsymbol{\theta})$ that depends on an unknown parameter vector $\boldsymbol{\theta}$ in a **parameter space** Θ. We use the same notation for probability mass functions (discrete Y) and densities (continuous Y). For example, $Y \sim N(\mu, \sigma^2)$ has density $f(y \mid \mu, \sigma^2) = \sigma^{-1}\phi((y-\mu)/\sigma)$ with $\phi(z) = (2\pi)^{-1/2}\exp(-z^2/2)$, $\mu \in \mathbb{R}$ and $\sigma^2 > 0$. A parametric model need not be a textbook distribution; it can be built for a specific application.

Definition 1.1 (Model)**.** A model for a random sample is the assumption that $Y_1, \ldots, Y_n$ are i.i.d. with a known density (or mass) function $f(y \mid \boldsymbol{\theta})$ and unknown parameter $\boldsymbol{\theta} \in \Theta$.

Definition 1.2 (Correct specification)**.** A model is **correctly specified** if there exists a unique $\boldsymbol{\theta}_0 \in \Theta$ such that $f(y \mid \boldsymbol{\theta}_0) = f(y)$, the true density. $\boldsymbol{\theta}_0$ is the **true parameter value**. Uniqueness means that no other $\boldsymbol{\theta} \in \Theta$ satisfies $f(y \mid \boldsymbol{\theta}) = f(y \mid \boldsymbol{\theta}_0)$. A model is **misspecified** if no $\boldsymbol{\theta} \in \Theta$ satisfies $f(y \mid \boldsymbol{\theta}) = f(y)$.

Example 1.1 (Correct and incorrect models)**.** Let the true density be $f(y) = 2e^{-2y}$, $y > 0$.

- The exponential model $f(y \mid \lambda) = \lambda^{-1}e^{-y/\lambda}$ is correctly specified with $\lambda_0 = 1/2$.
- The gamma model $f(y \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1}e^{-\beta y}$ is correctly specified with $(\alpha_0, \beta_0) = (1, 2)$.
- The log-normal model $f(y \mid \theta, \nu) = (2\pi\nu)^{-1/2}y^{-1}\exp\{-(\ln y - \theta)^2/(2\nu)\}$ is misspecified: no (θ, ν) reproduces $2e^{-2y}$.

Uniqueness matters. A two-component normal mixture $p\,\phi_{\sigma_1}(y-\mu_1) + (1-p)\,\phi_{\sigma_2}(y-\mu_2)$ contains the standard normal $\phi(y)$, but at many parameter values: $(p, 0, 1, 0, 1)$ for any p, $(1, 0, 1, \mu_2, \sigma_2^2)$ for any (μ_2, σ_2^2), and so on. The model contains the truth, but $\boldsymbol{\theta}_0$ is not identified, so it is not correctly specified in the sense of Definition 1.2.

1.2 The likelihood function and the MLE

The **likelihood** is the joint density of the sample, evaluated at the observed data and viewed as a function of $\boldsymbol{\theta}$. Under independence it factorises, and under identical distributions all factors share the same form:

$$L_n(\boldsymbol{\theta}) = f(y_1, \dots, y_n \mid \boldsymbol{\theta}) = \prod_{i=1}^{n} f(y_i \mid \boldsymbol{\theta}), \qquad \ell_n(\boldsymbol{\theta}) \equiv \ln L_n(\boldsymbol{\theta}) = \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{\theta}).$$

Definition 1.3 (Maximum likelihood estimator)**.** The MLE is the parameter value that maximises the likelihood,

$$\hat{\boldsymbol{\theta}}_n = \arg\max_{\boldsymbol{\theta} \in \Theta} L_n(\boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta} \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{\theta}).$$

The second equality holds because $\ln(\cdot)$ is strictly increasing and scaling by $1/n$ does not change the maximiser.

The MLE is the parameter value under which the observed sample is most likely. Working with the average log-likelihood is convenient: it is a sample mean, so laws of large numbers and central limit theorems apply directly.

Example 1.2 (Bernoulli)**.** Let $Y_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\pi)$, so $f(y \mid \pi) = \pi^y (1-\pi)^{1-y}$ for $y \in \{0, 1\}$. Then

$$\ell_n(\pi) = \sum_{i=1}^{n} \big[y_i \ln \pi + (1 - y_i) \ln(1 - \pi) \big] = n \big[\bar{y} \ln \pi + (1 - \bar{y}) \ln(1 - \pi) \big].$$

The first-order condition $n\bar{y}/\pi - n(1-\bar{y})/(1-\pi) = 0$ gives $\hat{\pi} = \bar{y}$. Figure 1.1 plots the likelihood for two samples of size five.

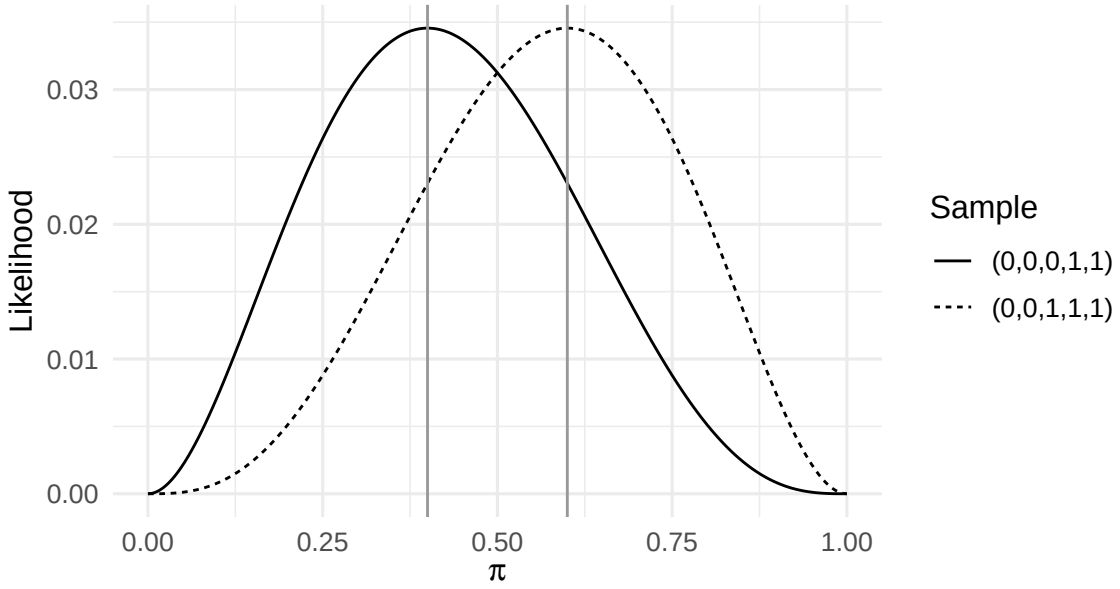


Figure 1.1: Bernoulli likelihood for the samples (0,0,0,1,1) and (0,0,1,1,1). The maximum is at the sample mean.

In practice we find the MLE in one of three ways:

1. **Solve the first-order condition** $\partial \ell_n(\boldsymbol{\theta})/\partial \boldsymbol{\theta} = \mathbf{0}$ and check the second-order condition. The FOC is necessary but not sufficient, and it only locates interior extrema; a maximum on the boundary of Θ need not satisfy it.
2. **Direct argument**, for instance by finding an upper bound of the likelihood that is attained at a unique point. The uniform distribution on $[0, \theta]$ is the standard case (Problem Set 1).
3. **Numerical optimisation** when no closed form exists (logit, probit, Tobit, ...). We then have to worry about convergence to a local rather than a global maximum.

Example 1.3 (Exponential)**.** Let $f(y \mid \lambda) = \lambda^{-1} e^{-y/\lambda}$, $y > 0$. The log-likelihood and its derivatives are

$$\ell_n(\lambda) = -n \ln \lambda - \frac{n\bar{y}}{\lambda}, \qquad \frac{d\ell_n}{d\lambda} = -\frac{n}{\lambda} + \frac{n\bar{y}}{\lambda^2}, \qquad \frac{d^2\ell_n}{d\lambda^2} = \frac{n}{\lambda^2} - \frac{2n\bar{y}}{\lambda^3}.$$

The unique root of the FOC is $\hat{\lambda} = \bar{y}$, and $d^2\ell_n(\hat{\lambda})/d\lambda^2 = -n/\bar{y}^2 < 0$ confirms a maximum.

Theorem 1.1 (Invariance)**.** *If $\hat{\boldsymbol{\theta}}$ is the MLE of $\boldsymbol{\theta} \in \mathbb{R}^m$, then for any function $\boldsymbol{\beta} = h(\boldsymbol{\theta}) \in \mathbb{R}^l$ the MLE of $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}} = h(\hat{\boldsymbol{\theta}})$.*

In Example 1.3, reparametrise with the rate $\beta = 1/\lambda$. Then $\ell_n(\beta) = n \ln \beta - \beta n \bar{y}$, maximised at $\hat{\beta} = 1/\bar{y} = h(\hat{\lambda})$, as the theorem predicts. Most estimators do not share this property; an unbiased estimator of λ, for example, does not deliver an unbiased estimator of $1/\lambda$.

1.3 Score, Hessian and information

Assume $f(y \mid \boldsymbol{\theta})$ is twice differentiable in $\boldsymbol{\theta} \in \mathbb{R}^k$.

Definition 1.4 (Score and Hessian)**.** The **score** is the $k \times 1$ gradient of the log-likelihood and the **Hessian** is the $k \times k$ matrix of second derivatives:

$$S_n(\boldsymbol{\theta}) = \frac{\partial \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \sum_{i=1}^{n} \underbrace{\frac{\partial \ln f(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}}_{S_i(\boldsymbol{\theta})}, \qquad H_n(\boldsymbol{\theta}) = \frac{\partial^2 \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'} = \sum_{i=1}^{n} \underbrace{\frac{\partial^2 \ln f(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'}}_{H_i(\boldsymbol{\theta})}.$$

The score measures how sensitive the log-likelihood is to the parameters; at an interior MLE, $S_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$. The Hessian is symmetric and measures curvature. If ℓ_n is strictly concave, H_n is negative definite and the maximiser is unique. Both are sums over observations because the log-likelihood is.

Theorem 1.2 (Properties of the likelihood)**.** *Suppose the model is correctly specified, the support of Y does not depend on $\boldsymbol{\theta}$, and $\boldsymbol{\theta}_0$ is in the interior of Θ. Then*

(i) $\mathbb{E}[S_i(\boldsymbol{\theta}_0)] = \mathbf{0}$*;*

(ii) ***information matrix equality****:* $\mathbb{E}[S_i(\boldsymbol{\theta}_0) S_i(\boldsymbol{\theta}_0)'] = -\mathbb{E}[H_i(\boldsymbol{\theta}_0)] \equiv \mathcal{I}_1(\boldsymbol{\theta}_0)$.

$\mathcal{I}_1$ is the (Fisher) information of a single observation and $\mathcal{I}(\boldsymbol{\theta}_0) = n\mathcal{I}_1(\boldsymbol{\theta}_0)$ that of the sample. Because the score has mean zero, the left side of (ii) is the variance of the score. Part (i) follows from differentiating $\int f(y \mid \boldsymbol{\theta})\, dy = 1$ under the integral sign; part (ii) from differentiating once more. The support condition guarantees that this exchange of differentiation and integration is allowed. The uniform $U[0, \theta]$ violates it, which is why its MLE behaves non-standardly.

Theorem 1.3 (Cramér–Rao lower bound)**.** *Under the same regularity conditions, any unbiased estimator $\tilde{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}_0$ satisfies* $\operatorname{Var}(\tilde{\boldsymbol{\theta}}) \succeq \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$.

Example 1.4 (Bernoulli, continued)**.** For one observation, $S_i(\pi) = \dfrac{y_i - \pi}{\pi(1-\pi)}$ and $H_i(\pi) = -\dfrac{y_i}{\pi^2} - \dfrac{1-y_i}{(1-\pi)^2}$. At π_0: $\mathbb{E}[S_i] = 0$, $\operatorname{Var}(S_i) = \dfrac{\pi_0(1-\pi_0)}{\pi_0^2(1-\pi_0)^2} = \dfrac{1}{\pi_0(1-\pi_0)}$, and $-\mathbb{E}[H_i] = \dfrac{1}{\pi_0} + \dfrac{1}{1-\pi_0} = \dfrac{1}{\pi_0(1-\pi_0)}$. The information equality holds and the CRLB is $\pi_0(1-\pi_0)/n$, which $\hat{\pi} = \bar{y}$ attains.

1.3.1 Conditional likelihood

Econometric models are usually about the distribution of y given covariates $\boldsymbol{x}$. Write the joint density as $f(y, \boldsymbol{x} \mid \boldsymbol{\delta}) = f(y \mid \boldsymbol{x}; \boldsymbol{\theta})\, f(\boldsymbol{x} \mid \boldsymbol{\gamma})$ with $\boldsymbol{\delta} = (\boldsymbol{\theta}', \boldsymbol{\gamma}')'$. If $\boldsymbol{\theta}$ and $\boldsymbol{\gamma}$ share no elements and are not linked by restrictions,

$$\ell_n(\boldsymbol{\delta}) = \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta}) + \sum_{i=1}^{n} \ln f(\boldsymbol{x}_i \mid \boldsymbol{\gamma}),$$

and $\boldsymbol{\theta}$ can be estimated from the **conditional log-likelihood** $\sum_i \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta})$ alone, leaving the distribution of $\boldsymbol{x}$ unspecified. All results above carry over with expectations taken over the joint distribution of $(y, \boldsymbol{x})$.

1.4 Asymptotic properties of the MLE

Theorem 1.4 (Consistency (Wooldridge 2010, Thm. 13.1))**.** *Suppose (a) the conditional density $f(y \mid \boldsymbol{x}; \boldsymbol{\theta})$ is correctly specified; (b) $\boldsymbol{\theta}_0$ is globally identified: for every $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$ in Θ, $P\{f(y \mid \boldsymbol{x}; \boldsymbol{\theta}) \neq f(y \mid \boldsymbol{x}; \boldsymbol{\theta}_0)\} > 0$; (c) Θ is compact and the log-likelihood is continuous in $\boldsymbol{\theta}$ with a dominance condition. Then $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$.*

The idea: $\frac{1}{n}\ell_n(\boldsymbol{\theta}) \xrightarrow{p} \mathbb{E}[\ln f(y \mid \boldsymbol{x}; \boldsymbol{\theta})]$ uniformly in $\boldsymbol{\theta}$, and the limit is uniquely maximised at $\boldsymbol{\theta}_0$ (this follows from the KLIC inequality in Section 1.6). The maximiser of the sample objective then converges to the maximiser of the limit.

Theorem 1.5 (Asymptotic normality and efficiency)**.** *Under additional regularity conditions (interior $\boldsymbol{\theta}_0$, twice continuously differentiable log-density, nonsingular $\mathcal{I}_1(\boldsymbol{\theta}_0)$),*

$$\sqrt{n}\,(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N\big(\mathbf{0},\, \mathcal{I}_1(\boldsymbol{\theta}_0)^{-1}\big).$$

The MLE is asymptotically efficient: its asymptotic variance attains the Cramér–Rao bound, and no consistent and asymptotically normal estimator has a smaller asymptotic variance under the same assumptions.

Sketch. A mean-value expansion of the FOC around $\boldsymbol{\theta}_0$ gives $\mathbf{0} = S_n(\hat{\boldsymbol{\theta}}) = S_n(\boldsymbol{\theta}_0) + H_n(\bar{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$, so

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \Big[-\tfrac{1}{n} H_n(\bar{\boldsymbol{\theta}})\Big]^{-1} \tfrac{1}{\sqrt{n}} S_n(\boldsymbol{\theta}_0).$$

By the LLN and consistency, $-\frac{1}{n}H_n(\bar{\boldsymbol{\theta}}) \xrightarrow{p} \mathcal{I}_1(\boldsymbol{\theta}_0) \equiv \boldsymbol{A}_0$. By the CLT and Theorem 1.2, $\frac{1}{\sqrt{n}} S_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{B}_0)$ with $\boldsymbol{B}_0 = \mathbb{E}[S_i S_i']$. Slutsky gives the general form $N(\mathbf{0}, \boldsymbol{A}_0^{-1}\boldsymbol{B}_0\boldsymbol{A}_0^{-1})$, which collapses to $\mathcal{I}_1^{-1}$ because $\boldsymbol{A}_0 = \boldsymbol{B}_0$ under correct specification. Keep this sandwich form in mind; it reappears in Section 1.6 and in GMM.

1.5 Estimating the variance matrix

In finite samples we use $\operatorname{Var}(\hat{\boldsymbol{\theta}}) \approx \frac{1}{n}\mathcal{I}_1(\boldsymbol{\theta}_0)^{-1} = \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$ and replace $\mathcal{I}_1$ by a consistent estimator. Three options are available, plus a fourth that does not rely on the information equality.

Table 1.1: Variance estimators after ML

	Estimator of $\operatorname{Var}(\hat{\boldsymbol{\theta}})$	Requires
(i) Expected Hessian	$\Big[-\sum_i \mathbb{E}\big(H_i(\boldsymbol{\theta}) \mid \boldsymbol{x}_i\big)\Big]^{-1}_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}$	closed form of $\mathbb{E}(H_i \mid \boldsymbol{x}_i)$
(ii) Observed Hessian	$\Big[-H_n(\hat{\boldsymbol{\theta}})\Big]^{-1}$	second derivatives (analytic or numerical)
(iii) Outer product of gradients (OPG, BHHH)	$\Big[\sum_i S_i(\hat{\boldsymbol{\theta}}) S_i(\hat{\boldsymbol{\theta}})'\Big]^{-1}$	first derivatives only
(iv) Robust (sandwich)	$\big[-H_n(\hat{\boldsymbol{\theta}})\big]^{-1}\Big[\sum_i S_i S_i'\Big]\big[-H_n(\hat{\boldsymbol{\theta}})\big]^{-1}$	both

(i)–(iii) are all consistent under correct specification and typically differ in finite samples. The observed Hessian (ii) is the default in most software. The OPG (iii) is the cheapest but has the poorest finite-sample behaviour. The sandwich (iv) remains valid when the density is misspecified, as we see next.

For the Bernoulli model, all three of (i)–(iii) coincide at $\hat{\pi} = \bar{y}$ and equal $\bar{y}(1-\bar{y})/n$. Verifying this is a short exercise: use $\sum_i (y_i - \bar{y})^2 = n\bar{y}(1-\bar{y})$ for binary y_i.

1.5.1 Example: the normal linear model

Let $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \mid \boldsymbol{x}_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Then

$$\ell_n(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2.$$

Setting the derivatives to zero yields

$$\hat{\boldsymbol{\beta}}_{ML} = \Big(\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\Big)^{-1}\sum_i \boldsymbol{x}_i y_i, \qquad \hat{\sigma}^2_{ML} = \frac{1}{n}\sum_i (y_i - \boldsymbol{x}_i'\hat{\boldsymbol{\beta}})^2.$$

$\hat{\boldsymbol{\beta}}_{ML}$ is OLS. $\hat{\sigma}^2_{ML}$ lacks the degrees-of-freedom correction, so it is biased downward in finite samples, but the bias vanishes as $n \to \infty$.

1.6 Misspecification and pseudo-ML

What does the MLE estimate when the model is wrong? The answer uses a measure of distance between densities.

Definition 1.5 (Kullback–Leibler divergence)**.** For densities f and g, $\operatorname{KLIC}(f, g) = \int f(y)\ln\dfrac{f(y)}{g(y)}\,dy$.

KLIC is not symmetric, $\mathrm{KLIC}(f,g) \neq \mathrm{KLIC}(g,f)$ in general, so it is not a distance in the metric sense. Its key properties are $\mathrm{KLIC}(f,f) = 0$, $\mathrm{KLIC}(f,g) \geq 0$ (by Jensen's inequality) and therefore $f = \arg\min_g \mathrm{KLIC}(f,g)$. Applied to a parametric family $f_{\boldsymbol{\theta}} = f(\cdot \mid \boldsymbol{\theta})$: if the model is correct, $\boldsymbol{\theta}_0 = \arg\min_{\boldsymbol{\theta}} \mathrm{KLIC}(f, f_{\boldsymbol{\theta}})$.

When the model is misspecified, the parametric family may still be a good **approximating model**, for example a log-normal density for wages. The natural target is then the best-fitting member of the family.

Definition 1.6 (Pseudo-true parameter)**.** $\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\Theta} \mathrm{KLIC}(f, f_{\boldsymbol{\theta}})$.

Since $\mathrm{KLIC}(f, f_{\boldsymbol{\theta}}) = \int f \ln f \, dy - \mathbb{E}[\ln f(y \mid \boldsymbol{\theta})]$ and only the second term depends on $\boldsymbol{\theta}$, $\boldsymbol{\theta}^*$ maximises $\mathbb{E}[\ln f(y \mid \boldsymbol{\theta})]$. The MLE maximises the sample analogue of this expectation. ML therefore has two roles: it estimates $\boldsymbol{\theta}_0$ when the model is correct and $\boldsymbol{\theta}^*$ when it is not. In the second case we call it the **pseudo-** or **quasi-MLE** (PMLE).

Theorem 1.6 (Asymptotic distribution of the PMLE (Huber 1967; White 1982))**.** *Under regularity conditions,* $\hat{\boldsymbol{\theta}}_{PMLE} \xrightarrow{p} \boldsymbol{\theta}^*$ *and*

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{PMLE} - \boldsymbol{\theta}^*) \xrightarrow{d} N(\mathbf{0},\ \boldsymbol{A}^{*-1}\boldsymbol{B}^*\boldsymbol{A}^{*-1}), \qquad \boldsymbol{A}^* = -\mathbb{E}[H_i(\boldsymbol{\theta}^*)], \quad \boldsymbol{B}^* = \mathbb{E}[S_i(\boldsymbol{\theta}^*)S_i(\boldsymbol{\theta}^*)'].$$

The information equality fails, so $\boldsymbol{A}^* \neq \boldsymbol{B}^*$ and only the sandwich estimator (iv) in Table 1.1 is consistent. In general, $\boldsymbol{\theta}^* \neq \boldsymbol{\theta}_0$ and the PMLE is inconsistent for the parameter we care about. There is one important exception.

1.6.1 The linear exponential family

A density belongs to the **linear exponential family** (LEF) if it can be written

$$f(y \mid \mu) = \exp\{a(\mu) + b(y) + c(\mu)\, y\}, \qquad \mathbb{E}[y] = \mu, \quad \mathrm{Var}(y) = [c'(\mu)]^{-1}.$$

Table 1.2: Leading members of the linear exponential family

Distribution	$a(\mu)$, $b(y)$, $c(\mu)$	$\mathbb{E}[y]$	$\mathrm{Var}(y)$
Normal (σ^2 known)	$-\frac{\mu^2}{2\sigma^2} - \frac{1}{2}\ln(2\pi\sigma^2)$, $-\frac{y^2}{2\sigma^2}$, $\frac{\mu}{\sigma^2}$	μ	σ^2
Bernoulli	$\ln(1-\mu)$, 0, $\ln\frac{\mu}{1-\mu}$	μ	$\mu(1-\mu)$
Exponential	$-\ln\mu$, 0, $-1/\mu$	μ	μ^2
Poisson	$-\mu$, $-\ln y!$, $\ln\mu$	μ	μ

Theorem 1.7 (Consistency of LEF-based PMLE (Gouriéroux, Monfort, and Trognon 1984))**.** *If the assumed density is LEF and the conditional mean* $\mathbb{E}[y \mid \boldsymbol{x}] = m(\boldsymbol{x}, \boldsymbol{\theta}_0)$ *is correctly specified, the PMLE of* $\boldsymbol{\theta}$ *is consistent, whatever the true distribution of* y *given* $\boldsymbol{x}$*.*

The reason is that the LEF score is proportional to $y - m(\boldsymbol{x}, \boldsymbol{\theta})$, so its expectation is zero at $\boldsymbol{\theta}_0$ as soon as the mean is right. The practical consequence is important: Poisson regression consistently estimates an exponential conditional mean for *any* nonnegative outcome, count or not, provided inference uses the sandwich variance. Computer tutorial 1 illustrates this with overdispersed count data.

1.7 Hypothesis testing: the trinity

We test $H_0 : \boldsymbol{h}(\boldsymbol{\theta}_0) = \mathbf{0}$ against $H_1 : \boldsymbol{h}(\boldsymbol{\theta}_0) \neq \mathbf{0}$, where $\boldsymbol{h} : \mathbb{R}^k \to \mathbb{R}^q$ imposes $q \leq k$ restrictions with full-rank Jacobian $\boldsymbol{R}(\boldsymbol{\theta}) = \partial \boldsymbol{h}(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}'$. Linear restrictions are the special case $\boldsymbol{h}(\boldsymbol{\theta}) = \boldsymbol{R}\boldsymbol{\theta} - \boldsymbol{r}$. For example, with $k = 3$, $H_0 : \theta_1 = 1,\ \theta_2 - \theta_3 = 2$ has $q = 2$ and

$$\boldsymbol{R} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \boldsymbol{r} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Let $\hat{\boldsymbol{\theta}}_U$ be the unrestricted MLE and $\hat{\boldsymbol{\theta}}_R$ the MLE subject to $\boldsymbol{h}(\boldsymbol{\theta}) = \mathbf{0}$. A restriction such as $\beta_1 = c\beta_2$ can often be imposed by reparametrisation: $x_1\beta_1 + x_2\beta_2 = (c\,x_1 + x_2)\beta_2$, so the restricted model has a single regressor.

Theorem 1.8 (LR, Wald and LM tests)**.** *Under H_0 and correct specification, each of the following converges in distribution to χ^2_q:*

$$\begin{aligned} LR &= 2\big[\ell_n(\hat{\boldsymbol{\theta}}_U) - \ell_n(\hat{\boldsymbol{\theta}}_R)\big], \\ W &= \boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)' \Big[\boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)\, \widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}_U)\, \boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)'\Big]^{-1} \boldsymbol{h}(\hat{\boldsymbol{\theta}}_U), \\ LM &= S_n(\hat{\boldsymbol{\theta}}_R)'\, \widehat{\mathrm{Var}}(S_n(\hat{\boldsymbol{\theta}}_R))^{-1} S_n(\hat{\boldsymbol{\theta}}_R), \qquad \widehat{\mathrm{Var}}(S_n(\hat{\boldsymbol{\theta}}_R)) = \mathcal{I}(\hat{\boldsymbol{\theta}}_R) \text{ or } -H_n(\hat{\boldsymbol{\theta}}_R). \end{aligned}$$

The three statistics are asymptotically equivalent under H_0 and under local alternatives.

The intuition differs (Figure 1.2):

- **LR**: if the restriction is valid, imposing it should not reduce the maximised log-likelihood by much.
- **Wald**: if the restriction is valid, $\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)$ should be close to zero, because $\hat{\boldsymbol{\theta}}_U$ is consistent.
- **LM (score)**: if the restriction is valid, the slope of the log-likelihood at $\hat{\boldsymbol{\theta}}_R$ should be close to zero, because $\hat{\boldsymbol{\theta}}_R$ is then close to the unrestricted maximum.

For a single coefficient, $H_0 : \theta_j = 0$, the Wald statistic is the square of the usual t-ratio $\hat{\theta}_j / \mathrm{se}(\hat{\theta}_j)$.

The three statistics for this sample:

```
# y: 50 draws from an exponential with lambda = 1.5 (set.seed(42)); H0: lambda
↪  = 1.2
info <- function(l) n / l^2                  # Fisher information of the sample
LR <- 2 * (ll(l_hat) - ll(l0))
W  <- (l_hat - l0)^2 * info(l_hat)           # uses Var(l_hat) = l_hat^2 / n
LM <- slope^2 / info(l0)                     # score and information at l0
round(c(LR = LR, Wald = W, LM = LM, crit_5pct = qchisq(0.95, 1)), 3)
```

```
#>        LR      Wald        LM crit_5pct
#>     6.907     4.361     8.782     3.841
```

They differ in finite samples but lead to the same decision here. Which to use in practice?

1. **Computation.** LR needs both models, Wald only the unrestricted one, LM only the restricted one. LM is attractive when the restricted model is much simpler, e.g. when the restriction makes a nonlinear model linear.

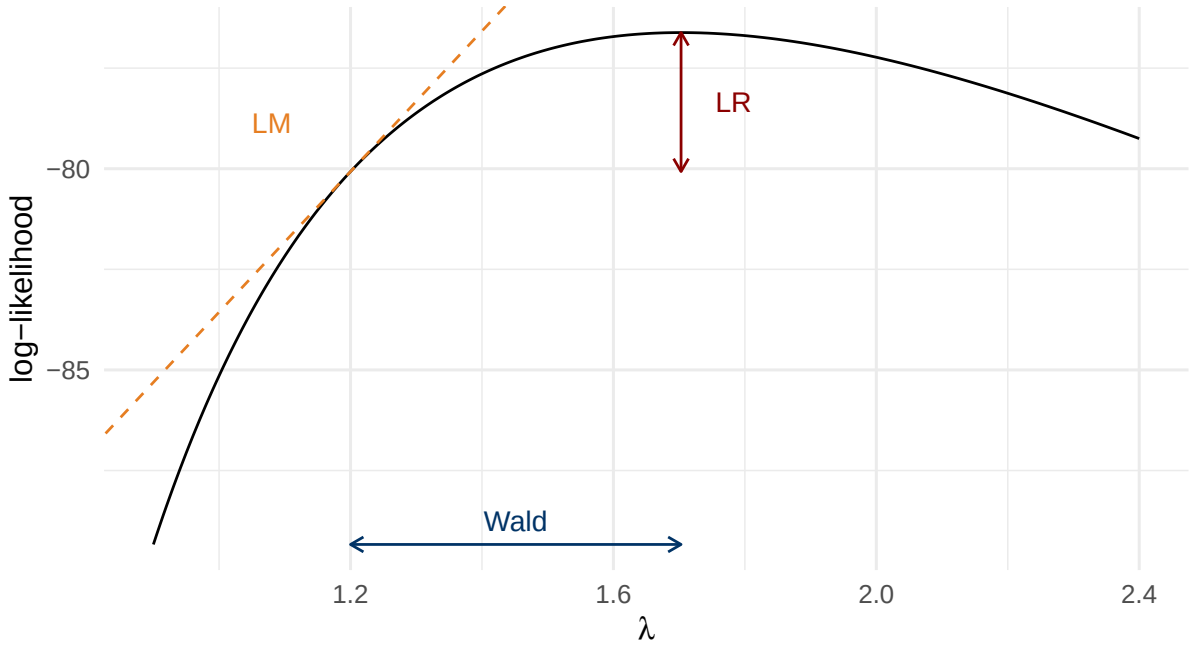


Figure 1.2: The trinity for the exponential model, testing $H_0 : \lambda = 1.2$ with n = 50 simulated observations. LR compares heights, Wald compares horizontal distance, LM compares the slope at the restricted value.

2. **Invariance.** Only LR is invariant to how the model and the restriction are parametrised. The Wald statistic for $H_0 : \theta = 1$ and for the equivalent $H_0 : 1/\theta = 1$ can differ substantially in small samples (Cameron and Trivedi 2005, sec. 7.2).
3. **Robustness.** Under misspecification, LR and the information-based LM are no longer χ^2. The Wald test with the sandwich variance remains valid, which is why robust Wald tests dominate applied work.

1.8 Computer tutorial 1

Learning goals. Write a log-likelihood as an R function and maximise it with `optim`; compute Hessian, OPG and sandwich variance estimators by hand and check them against `glm` and `sandwich`; see in a Monte Carlo study why pseudo-ML needs robust standard errors.

Packages. `sandwich`, `lmtest`, `MASS`. **Data.** `mroz.csv` (Mroz 1987), 753 married women in 1975.

1.8.1 Problem 1: Logit model for labour force participation

Variable	Description
`inlf`	= 1 if in labour force, 1975
`hours`	hours worked, 1975
`kidslt6`	number of kids under 6
`kidsge6`	number of kids aged 6–18
`age`	woman's age in years
`educ`	years of schooling

Variable	Description
`wage`	estimated hourly wage from earnings (0 if not working)
`faminc`	family income, 1975
`exper`	actual labour market experience
`expersq`	experience squared

a) Create non-wife income in thousand dollars, `nwifeinc` = (`faminc` − `wage` × `hours`)/1000.

b) Estimate the linear probability model (LPM)

$$\texttt{inlf} = \beta_0 + \beta_1 \texttt{nwifeinc} + \beta_2 \texttt{educ} + \beta_3 \texttt{exper} + \beta_4 \texttt{expersq} + \beta_5 \texttt{age} + \beta_6 \texttt{kidslt6} + \beta_7 \texttt{kidsge6} + \varepsilon.$$

Why are heteroskedasticity-robust standard errors required here? How many fitted values lie outside $[0, 1]$?

c) Write down the logit model and its log-likelihood formally. Derive the score and the Hessian of observation i.

d) Estimate the logit model with the regressors from (b) in two ways.

1. Code the **negative** log-likelihood as an R function `negll(b, y, X)` and minimise it with `optim(..., method = "BFGS", hessian = TRUE)`, using the LPM coefficients as starting values. Supply the analytical gradient as well.
2. Use `glm(..., family = binomial(link = "logit"))`.

e) Compute the variance matrix of $\hat{\boldsymbol{\beta}}$ in four ways: (i) from the Hessian returned by `optim`; (ii) the inverse of the negative analytical Hessian; (iii) the inverse of the OPG; (iv) the sandwich. Compare with `vcov()` and `sandwich()` applied to the `glm` object.

f) Compute the average partial effects (APEs) of `educ` and `kidslt6` from the logit model and compare them with the LPM coefficients.

1.8.2 Problem 2: PML vs. ML, a Monte Carlo study

The goal is to see how ML behaves when the likelihood is misspecified but the conditional mean is right (see "Misspecification and pseudo-ML" in Chapter 1 of the lecture notes).

Correct density.

a) Simulate $n = 1000$ observations from $y_i \mid \boldsymbol{x}_i \sim \text{Poisson}(\lambda_i)$ with $\lambda_i = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})$, $(\beta_0, \beta_1, \beta_2) = (0.2, 0.1, 0)$ and x_1, x_2 independent $N(0, 1)$.

b) Estimate $\boldsymbol{\beta}$ with `glm(..., family = poisson)`.

c) Using the score and Hessian of the Poisson model from Problem Set 1, Problem 3, compute Hessian-based and sandwich standard errors by hand. Check them against `glm` and `coeftest(..., vcov = sandwich)`.

Misspecified density.

d) Now draw y_i from a negative binomial distribution with the same mean $\mu_i = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})$ and variance $\text{Var}(y_i \mid \boldsymbol{x}_i) = \mu_i + \mu_i^2/\theta$ with $\theta = 3$.

e) Estimate the mean function by (i) Poisson PML with Hessian-based variance, (ii) Poisson PML with sandwich variance, (iii) negative binomial ML (`MASS::glm.nb`) with Hessian-based variance.

f) Repeat (d)–(e) for 1000 Monte Carlo samples and fill in the table:

	Poisson PML (Hessian)	Poisson PML (sandwich)	NegBin ML (Hessian)
MC mean of $\hat{\beta}_2$			
MC mean of s.e.($\hat{\beta}_2$)			
MC std. dev. of $\hat{\beta}_2$			
Rejection rate of $H_0 : \beta_2 = 0$ at 5%			

g) Discuss your results.

1.9 Exercises

Problem Set 1 contains the analytical exercises for this chapter:

- **Problem 1**: a regular one-parameter model from start to finish (score, information, bias and the Cramér–Rao bound, asymptotics, invariance) and the three tests, including the non-invariance of the Wald test.
- **Problem 2**: the uniform distribution as a non-regular case, with an n-rate, non-normal limit.
- **Problem 3**: Poisson regression, its score and Hessian, and pseudo-ML under overdispersion. Computer tutorial 1, Problem 2 builds on it.
- **Problem 4**: canonical links in the linear exponential family.

2 Method of Moments and GMM

In preparation

This chapter will be added before the corresponding lectures. Planned content:

- Moment conditions, MM and GMM estimators
- Identification, consistency and asymptotic normality
- Optimal weighting matrix, two-step and iterated GMM
- IV and 2SLS as GMM; overidentification (J) test
- Example: ATE by inverse probability weighting as a joint GMM problem
- Computer tutorial 2

Part II

Computation and flexible models

3 Simulation-Based Estimation

> **In preparation**
>
> This chapter will be added before the corresponding lectures. Planned content:
>
> - Monte Carlo integration and its error
> - Quasi-random (Halton) draws and Gaussian quadrature
> - Maximum simulated likelihood and method of simulated moments
> - Bootstrap
> - Computer tutorial 3

4 Nonparametric Methods

In preparation

This chapter will be added before the corresponding lectures. Planned content:

- Histogram and kernel density estimation; bias, variance and bandwidth choice
- Nadaraya-Watson and local linear regression
- Rule-of-thumb and cross-validated bandwidths; confidence bands
- Curse of dimensionality and semiparametric alternatives
- Computer tutorial 4

5 Introduction to Statistical Learning

In preparation

This chapter will be added before the corresponding lectures. Planned content:

- Prediction vs. estimation; bias-variance trade-off
- Cross-validation
- Ridge, lasso and elastic net
- Trees and ensembles (overview)
- Lasso for inference and double machine learning (outlook)
- Computer tutorial 5

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A Large-Sample Asymptotics

> **In preparation**
>
> This chapter will be added before the corresponding lectures. Planned content:
>
> - Convergence in probability and distribution; Slutsky and continuous mapping
> - WLLN, CLT; delta method
> - Uniform convergence and consistency of extremum estimators

B A Short R Primer

> **In preparation**
>
> This chapter will be added before the corresponding lectures. Planned content:
>
> - Projects, scripts and working directories
> - Data import, data frames and matrices
> - Matrix algebra and OLS by hand
> - Writing functions; optimisation with optim
> - Computer tutorial 0